

**DEVELOPMENT OF AN AGENTIC CYBER-PHYSICAL SYSTEM FOR
AUTONOMOUS AND SUSTAINABLE INDOOR PRECISION
AGRICULTURE IN NEPAL**

Submitted By

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Date of Submission: June 1, 2026

Submitted To

Nepal Academy of Science and Technology (NAST)

Khumaltar, Lalitpur, Nepal

ABSTRACT

Farmers in Nepal face serious problems from climate change, irregular weather, and rising costs. Traditional farming and standard plastic greenhouses struggle to keep stable growing conditions. Simple timer-based watering systems run on fixed schedules and don't respond to sudden changes in temperature or humidity. This causes plant stress, wasted water, and lower yields. To fix these problems, this research builds and tests a low-cost, closed-loop Cyber-Physical System (CPS) that uses an AI agent to automate daily plant care.

The system uses affordable sensors to measure air temperature, humidity, leaf temperature, soil moisture, pH, EC, and water level. These sensors connect directly to a localized network of ESP32 microcontrollers. A central Raspberry Pi edge gateway stores the collected data and manages communication. A Large Language Model (LLM) acts as the main decision maker. It reads real-time sensor data, calculates Vapor Pressure Deficit (VPD) and Daily Light Integral (DLI), and sends safe commands to pumps, lights, fans, and nutrient dosers.

For safety, we add hardware, firmware, and software backups to stop mistakes or crashes. Over 12 months, we will build, calibrate, and test the system by growing three batches of radish microgreens. We will check whether this AI-driven system saves water and nutrients better than standard timers. This will give Nepal a practical model for smart, resource-saving farming.

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1 Introduction

Nepali farmers struggle with unpredictable weather, late rains, and sudden temperature changes. Open fields and simple plastic greenhouses don't protect crops well enough. Basic timer-based systems water on a fixed schedule, even when the soil is already wet or dry. This fixed schedule causes diseases, wastes water and nutrients, and lowers harvest quality.

To solve these issues, we propose the development of an affordable, safe, and automated growing system. By connecting simple microcontrollers to an intelligent AI program, the system creates a smart loop that constantly monitors the growing space and makes real-time adjustments on its own. This approach will help urban and semi-urban farmers save water, reduce physical labor, and maintain steady crop yields even during extreme weather conditions.

1.1 Research Problem

Farming in Nepal is very sensitive to changing climate. To grow food reliably, urban and semi-urban growers are starting to use indoor farms and small greenhouses. But high-value crops like microgreens need very steady temperature, humidity, and soil moisture.

Manual tracking is time-consuming and prone to human error, while standard automatic systems rely on rigid timers. These simple timers cannot manage multiple conditions at once, which can reduce crop yields by 30% to 50% due to under-watering or over-watering. On the other hand, advanced commercial smart greenhouses are too expensive for small-scale growers and research institutions in Nepal, and they lack local technical support. There is a practical need for an affordable, open-source, and highly secure automated system that can safely control water, light, and nutrients on its own.

1.2 Research Objectives

The primary goal of this research is to build and test a low-cost, safe, and autonomous system for indoor precision agriculture. The specific objectives are:

- To design and build a modular, low-voltage Ebb and Flow precision cultivation system.
- To set up a local edge network using ESP32 microcontrollers to send data and commands in real time.
- To integrate an intelligent Large Language Model (LLM) agent that reads environmental data and safeguards growing conditions.
- To design and test multi-layered, fail-safe safety rules across the physical, firmware, and software layers to block any incorrect or unsafe commands.
- To measure and compare the water, nutrient, and energy efficiency of this AI-driven system against traditional timer-based automation across three complete crop cycles.

2 Literature Review

Connecting physical farming systems with computer programs has laid the groundwork for modern smart agriculture. Early precision farming relied on simple remote monitoring and automated controls. Recently, researchers have used machine learning and cloud databases to build more advanced "cyber agriculture" setups (Johnson et al., 2019). Studies show that crop growth can be optimized by treating conditions like temperature and light as dynamic variables that can be adjusted in real time (Johnson et al., 2019).

However, simple automated systems are limited because they follow rigid rules. Gangurde et al. (2025) note that while basic automated systems help collect data, they struggle to adapt when weather conditions change quickly. For example, a simple system cannot easily cool down a greenhouse without accidentally making the air too dry (Gangurde et al., 2025).

To solve this, researchers are studying "Agentic" Artificial Intelligence (AAI) for precision farming (Srinivasu et al., 2025). Unlike basic programs that only follow simple rules, an intelligent AI agent can look at multiple sensor readings at once and make smart decisions to keep conditions stable (Srinivasu et al., 2025). This helps conserve resources like water and power while keeping plants healthy.

Recent projects have shown that Large Language Models (LLMs) can be used as the central decision-maker in these setups (Murad et al., 2026). The LLM reads the sensor data, reasons about what the plants need, and sends commands to physical components like fans and pumps (Murad et al., 2026). Additionally, using camera-based monitoring, as shown in the Greenhouse-5.0 project, provides a non-invasive way to track crop growth and health (Di Caro, 2022).

This project builds on these studies to construct an affordable, safe, and automated farming system. The design integrates real-time sensors and edge computing with strict, multi-layered safety rules to ensure the system remains safe and reliable under local conditions in Nepal.

3 Materials and Methods

The proposed research will be conducted at a designated laboratory facility in Lalitpur, Nepal. The methodology is structured across physical system design, embedded hardware configuration, software orchestration, safety engineering, and empirical testing.

3.1 Physical and Embedded Architecture

The physical cultivation system will utilize a robust Ebb and Flow (flood and drain) hydroponic system. The system's architecture is organized into five operational layers, as detailed in Figure 1.

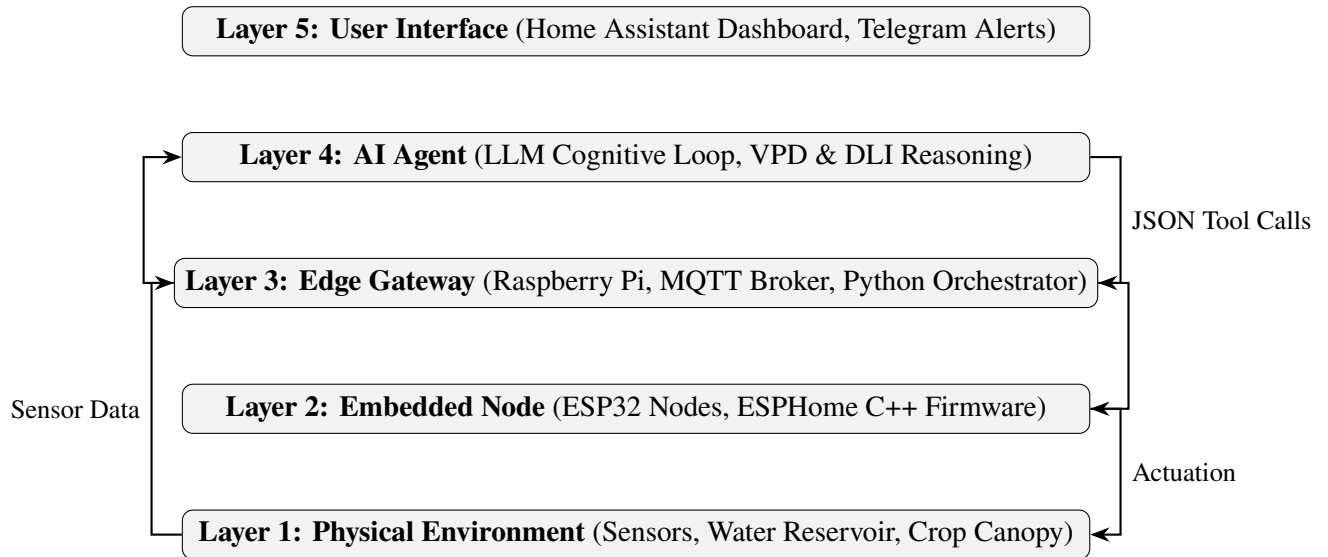


Figure 1. Layered Cyber-Physical Architecture Diagram

The five integrated layers consist of:

1. **Physical Layer:** Comprises the physical structure, including an opaque 100-liter water reservoir, cultivation trays, support structures, and the physical crop canopy.
2. **Embedded Layer:** Consists of localized ESP32 microcontrollers configured using ESPHome, compiling highly stable C++ code directly on-chip to handle sensor reading and localized actuation.
3. **Edge Gateway Layer:** Uses a local Raspberry Pi 4 server hosting an MQTT broker (Eclipse Mosquitto) for low-latency communication, an InfluxDB time-series database for logging, and a custom Python-based FastAPI Orchestrator.
4. **AI Agent Layer:** Consists of a cloud-connected LLM engine. The agent periodically

requests the current environmental state vector, reasons about optimal conditions, and generates formatted JSON tool calls.

5. **User Interface Layer:** Provides real-time data visualization and manual control capabilities using a centralized Home Assistant dashboard and automated Telegram alerts.

Table 1. System Sensor Configuration Matrix

Sensor Type	Model Used	Interface	Operational Placement	Accuracy
Air Temp & Hum.	Sensirion SHT40	I2C	Plant canopy level	$\pm 0.1^{\circ}\text{C}$, $\pm 2\%$ RH
Leaf Temperature	MLX90614 (IR)	I2C	Pointed at crop canopy	$\pm 0.5^{\circ}\text{C}$
Soil Moisture	Capacitive v2.0	Analog	Root substrate	$\pm 3\%$ VWC
Water pH Level	Atlas EZO-pH	UART	Inside liquid reservoir	± 0.02 pH
Elec. Cond.	Atlas EZO-EC	UART	Inside liquid reservoir	$\pm 5 \mu\text{S}/\text{cm}$
Liquid Level	JSN-SR04T	GPIO	Top of reservoir tank	± 1 cm
Leak Detection	Resistive Strip	Digital	Protective floor	Binary (On/Off)

Table 2. System Actuator Configuration Matrix

Actuator Type	Power Spec.	Control Interface	Primary Purpose
Submersible Pump	12V DC, 2.0 A	Relay	Flooding the crop root zone
Solenoid Valve	12V DC, 0.8 A	Relay	Flow control & drain prevention
Grow Lights	120V AC, 100W	Relay / DAC	PAR light delivery (DLI control)
Exhaust Fan	12V DC, 0.5 A	PWM via MOSFET	Temp and humidity extraction
Dosing Pumps	12V DC, 0.5 A	Relays	Precision pH & nutrient adjustments

3.2 Safety and Fail-Safe Architecture

To ensure complete operational safety and prevent damage to physical equipment or crops due to network issues or software errors, a multi-layered fail-safe architecture is implemented, as outlined in Figure 2.

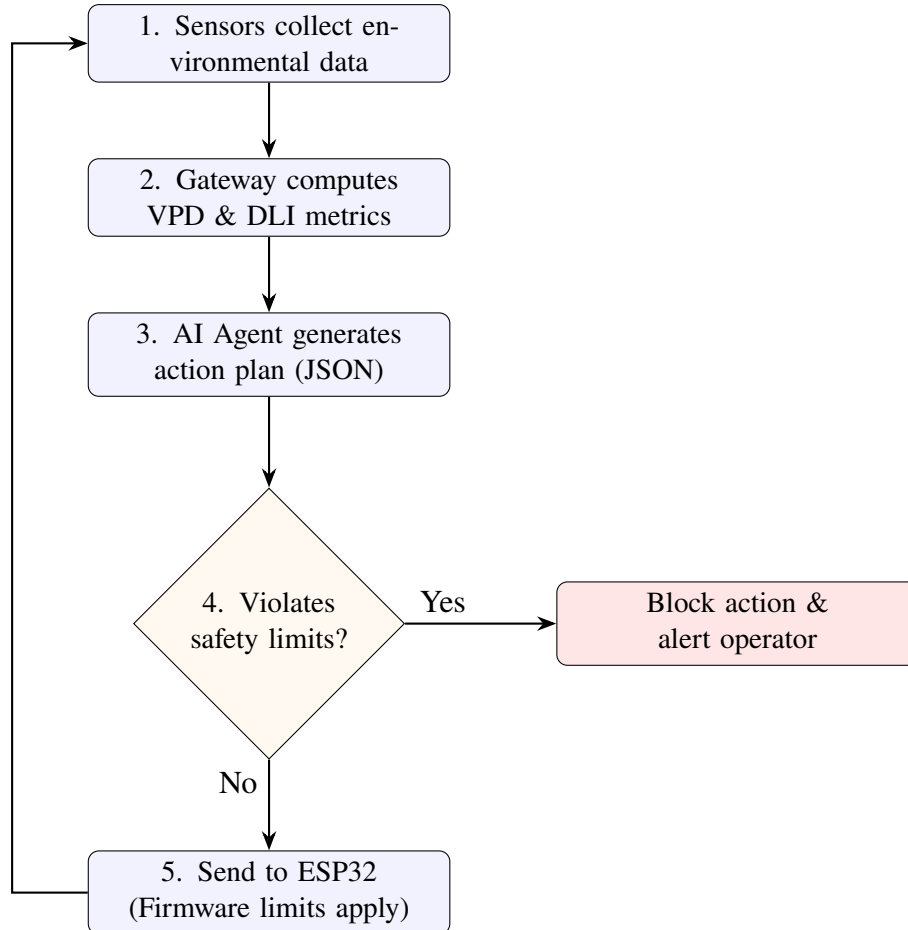


Figure 2. Closed-Loop Control Flow with Active Safety Gates

The integrated fail-safe boundaries operate on three distinct levels:

- **Layer 0 (Hardware/Physical):** A physical thermal fuse (rated at 80°C) is wired in series with the heater supply to physically cut power in case of thermal runaway. Mechanical float switches are placed in the reservoir to physically disconnect the pump's power circuit if the water level drops below a safe minimum.
- **Layer 1 (Firmware/Microcontroller):** The ESP32 firmware features hard-coded maximum runtime limits for physical actuators (e.g., the water pump is restricted to a maximum continuous run of 60 seconds). A hardware watchdog timer is enabled to automatically reboot the ESP32 and switch all relays to a default "OFF" state if the software loop hangs.
- **Layer 2 (Edge Gateway/Orchestrator):** The custom Python orchestrator uses a JSON-schema validator to inspect every tool command sent by the AI agent. If a command violates preset safety limits, the command is immediately blocked, the event is logged,

and an emergency alert is sent to the operator via Telegram.

3.3 Budget Breakdown

The total budget for this project is NRs.7,00,000 (Seven Lakhs). Out of this total, We plan to request a research grant of NRs.5,00,000 (Five Lakhs) from NAST. We will look for other supplementary grants or sponsorships to cover the remaining amount.

Table 3. Detailed Project Budget Breakdown (NRs.)

S.N.	Expenditure Details and Classifications	Cost (NRs.)
1	Instrument, Machinery and Tools Handheld calibration meters (pH/EC), basic electronic soldering stations, digital multi-meters, and diagnostic tools (limited to 5% of total budget).	35,000
2	Testing Soil/water chemical analysis, plant metabolite profiling at diagnostic labs, and software simulation validation tests.	1,00,000
3	Special Machining Laser cutting of custom acrylic panels, CNC milling for watertight custom enclosures, and mounting bracket fabrication.	70,000
4	Software API token credits for real-time agentic reasoning, cloud database hosting, secure MQTT broker, and system backup tools.	70,000
5	Construction Materials Extruded aluminum frames, grow trays, pH/EC/SHT40 sensors, ESP32 boards, LED grow lights, relays, solenoids, water pumps, pipes, and electrical safety components (GFCI, fuses).	2,20,000
6	Chemicals Hydroponic liquid nutrient stock solutions (A and B), pH calibrators, sensor calibration standards, and organic bio-fungicides.	60,000
7	Field Visit Local travel to greenhouses in Lalitpur and Bhaktapur for baseline data, transport of materials, and regular physical inspection (limited to 10% of total budget).	70,000
8	Remuneration Remuneration for a part-time research assistant or technician to help with daily seeding, harvesting, and system upkeep (limited to 10% of total budget).	50,000
9	Miscellaneous Printing, report preparation, dissemination of results, internet data packages, and small administrative overheads (limited to 5% of total budget).	25,000
Total Budget		7,00,000

4 Research Plan

The proposed research will be executed over a strict 12-month period. The project is structured chronologically into distinct milestones to guarantee successful delivery.

Table 4. Chronological Project Implementation Schedule

Phase	Key Research Activities and Operational Tasks	Timeline
Phase 1	Buy electrical parts, sensors, actuators, frame materials, and calibration liquids.	Months 1–2
Phase 2	Build the physical frame, connect the water pipes, and solder custom circuits.	Month 3
Phase 3	Program the ESP32 microcontroller with ESPHome and test local sensors and relays.	Month 4
Phase 4	Set up the Raspberry Pi, configure Home Assistant, database logging, and MQTT.	Months 5–6
Phase 5	Set up the AI agent, write the Python validation code, and test the safety loops.	Month 7
Phase 6	Test the system with water and nutrients, calibrate the sensors, and test fail-safes.	Month 8
Phase 7	Grow three full batches of radish microgreens under AI care and collect data.	Months 9–10
Phase 8	Analyze the collected data, compare resource use, and write the final NAST report.	Months 11–12

5 Expected Output and Conclusion

The implementation of this project is expected to deliver several key technical, scientific, and practical outputs:

- **Functional Prototype:** A fully operational, open-source Agentic Cyber-Physical greenhouse system designed for high safety and resource efficiency in local Nepalese contexts.
- **Software Repository:** A complete, well-documented codebase featuring ESP32 ES-Phome configurations, Python orchestrator scripts, and the JSON-schema safety validation rules, published on an open-source platform.
- **High-Fidelity Dataset:** A comprehensive, high-frequency time-series dataset capturing air temperature, leaf temperature, relative humidity, soil moisture, pH, EC, water levels, and corresponding AI action logs over three complete crop cycles.
- **Resource Optimization:** Concrete empirical evidence demonstrating a target 20% increase in water-use and nutrient efficiency, and stable microclimatic control (VPD 0.8-1.2 kPa) compared to traditional automation models.
- **Scientific Dissemination:** At least one peer-reviewed research paper submitted to a reputable national or international journal, and a comprehensive final project report presented to NAST.

In conclusion, this project will show that using a local edge computer and an AI agent together can reliably automate farming. By combining AI control with simple hardware and firmware backups, the system greatly reduces the risk of software mistakes or network failures. The result is a low-cost, practical method for data-driven farming that works well in Nepal.

6 References

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